Finance Research Letters 104 (2026) 110210



Contents lists available at ScienceDirect

Finance Research Letters

journal homepage: www.elsevier.com/locate/frl





AI washing: How does it influence firm value?

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ARTICLE INFO

Keywords:
AI washing
Firm value
Expectation confirmation
Expectation violation

ABSTRACT

This paper examines the effect of AI washing on firm value. Using data from Chinese listed companies from 2016 to 2024, we find a significant inverted U-shaped relationship between AI washing and firm value. Furthermore, this relationship is more pronounced in large, state-owned, and high-tech enterprises. This study not only reconciles previously conflicting views on the market response to AI washing but also offers valuable practical implications for managers, investors, and regulators.

1. Introduction

The rise of promising technology is usually accompanied by the emergence of firm opportunistic behavior. For example, during the boom of the internet industry, many firms tacked ".com" onto their names to attract investors (Cooper et al., 2001), only to drop the suffix after the dot-com bubble burst (Cooper et al., 2005). A similar form of opportunistic behavior is now emerging in the field of AI. Recent studies suggest that some firms deliberately exaggerate their AI capabilities, a phenomenon termed AI washing (Xu et al., 2026). Like greenwashing, AI washing stems from information asymmetry and represents a gap between firm claims and actual actions This industry-specific high (Sun et al., 2026).

However, from the perspective of signal distortion, greenwashing and AI washing differ fundamentally. Greenwashing distorts responsibility-related signals and thus affects evaluations of legitimacy and compliance. AI washing distorts capability-related signals and is more likely to reshape investors' valuation expectations regarding firms' future growth and competitive advantage. Moreover, firms' environmental performance can often be verified through relatively clear standards, such as emissions data and compliance reviews. AI capabilities, by contrast, are inherently difficult to verify, a problem compounded by the rapid iteration of AI technology. These differences make AI washing tend to be more prevalent and damaging than greenwashing.

Extant research has explored market responses to AI washing, yet no consensus exists. Some studies suggest that AI washing elicits positive market reactions, arguing that it conveys AI-related capability signals to the market, thereby increasing the market's valuation of firms' future growth potential (Liu and Li, 2026; Xing et al., 2026). Other studies, however, suggest that AI washing provokes negative market reactions, because willful, unfettered AI washing may erode trust among firms and external stakeholders such as investors (Sharma et al., 2025; Song et al., 2026; Liu et al., 2026). These mixed findings indicate that the market response to AI washing

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https://doi.org/10.1016/j.frl.2026.110210
Received 13 January 2026; Received in revised form 14 May 2026; Accepted 18 May 2026
Available online 19 May 2026
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is not yet well understood.

To reconcile these conflicting views, we argue that the market reaction to AI washing is likely to be nonlinear. Moderate AI washing allows firms to capture investor attention and ride market enthusiasm for AI-related growth, yet excessive AI washing triggers skepticism about firms' claims and ultimately backfires. Using data from Chinese listed companies from 2016 to 2024, we find support for our theoretical arguments, confirming the presence of a curvilinear relationship (i.e., an inverted U-shaped relationship). Furthermore, this effect is more pronounced in large, state-owned, and high-tech enterprises.

The contributions are as follows: First, existing research has predominantly focused on the linear relationship between AI washing and market response. Our study reconciles these conflicting views by incorporating a nonlinear relationship. Furthermore, the competitive displacement that elucidates why low-level AI washing reduces firm value enriches existing theoretical explanations. Second, existing research on firm washing has focused primarily on responsibility-related signals (e.g., greenwashing), whereas this study examines AI washing as a distortion of technological capability signals, thereby deepening the understanding of washing types.

2. Theoretical analysis and research hypotheses

The strong market enthusiasm for AI motivates many firms to engage in AI washing to attract market attention. Against this background, if a firm engages in only a low level of AI washing, it may not only fail to attract new investor attention but also risk losing its existing investor base and market visibility to firms with more aggressive AI washing, resulting in a competitive displacement effect and ultimately harming firm value. This stems not only from information asymmetry but also from the fact that investors typically compare peer firms and vote with their feet.

By contrast, a moderate level of AI washing can enhance a firm's visibility while generating expectation confirmation among investors and other stakeholders, thereby contributing to firm value. Expectation confirmation theory posits that when individuals' expectations are met, confirmation elicits positive emotions and optimistic judgments (Qi et al., 2026). We argue that moderate AI washing conveys more information but is insufficient to trigger external stakeholders' verification of its authenticity. Given the strong market enthusiasm for AI, this information largely aligns with stakeholders' expectations of the firm's AI development, thereby resulting in expectation confirmation. The positive emotions generated by expectation confirmation can drive collective actions (Huang et al., 2026), such as continued stock purchases, participation in equity offerings, more favorable credit conditions, and analyst upgrades, which in turn enhance firm value.

However, excessive AI washing may invite scrutiny of firms' actual AI initiatives and provoke expectation violations, thereby harming firm value. Expectation violation theory suggests that deviations between substantive actions and expectations trigger cognitive dissonance, leading to disappointment, dissatisfaction, or even anger (Yao and Xi, 2024). Following this logic, excessive AI washing may raise suspicion among key stakeholders and prompt verification of firms' AI claims through multiple channels. Investors may scrutinize AI-related investments disclosed in financial reports, while the media may intensify coverage of inconsistencies in AI implementation. When substantial discrepancies arise between symbolic claims and substantive actions, stakeholders experience strong expectation violations. This, in turn, incites negative reactions, including stock sell-offs, credit rating downgrades, and regulatory penalties (Eklund and Mannor, 2026), thus reducing firm value.

Taken together, we propose the following hypothesis:

H1: There is a significant inverted U-shaped relationship between AI washing and firm value.

We argue that the above effect will be more pronounced in large enterprises. Large enterprises possess resource advantages (Liu and Pan, 2026), leading stakeholders to harbor higher expectations for their AI transformation. While a low level of AI washing by large enterprises may struggle to meet stakeholders' expectations in a crowded market, moderate AI washing is more likely to generate expectation confirmation among stakeholders. However, it also means that the expectation violation triggered by excessive AI washing will be more severe. Stakeholders believe that large enterprises have the capacity and resources to fulfill their commitments; therefore, failure to do so leads to more severe reputational penalties and trust crises.

H2: The inverted U-shaped relationship between AI washing and firm value is more pronounced in large enterprises.

Consistent with the above logic, the inherent institutional identity of state-owned enterprises (SOEs) leads the market to hold higher expectations and trust in their implementation of national AI strategies. Moderate AI washing by SOEs is thus more likely to resonate with stakeholders' existing optimism and generate expectation confirmation. However, the expectation violation triggered by excessive AI washing will also be more severe. This will not only severely damage stakeholder trust but may also be regarded as a failure to fulfill strategic missions, leading to harsher reputational penalties and regulatory scrutiny.

H3: The inverted U-shaped relationship between AI washing and firm value is more pronounced in SOEs.

A similar logic applies to high-tech enterprises. Stakeholders expect these firms to be at the forefront of AI development. This industry-specific high expectations makes moderate AI washing credible (expectation confirmation). However, stakeholders also consider high-tech firms to have the capability to translate promises into actual outcomes. Therefore, a significant discrepancy between rhetoric and reality is less tolerated and is viewed as a failure of innovation management.

H4: The inverted U-shaped relationship between AI washing and firm value is more pronounced in high-tech enterprises.

3. Research design

3.1. Data sources

This paper uses Chinese listed firms from 2016 to 2024 as the research sample. Relevant data are downloaded from the CSMAR

database. The sample is standardized as follows: first, exclude ST and PT samples; second, exclude samples with missing key variables; finally, 25,334 observations are obtained. In addition, all continuous variables are winsorized at 1% and 99%.

3.2. Variable definitions

3.2.1. Dependent variable: firm value (TQ)

This paper measures firm value using Tobin's Q. The reason is that Tobin's Q is determined by the capital market and reflects investors' collective expectations of enterprises' development potential (Yang et al., 2025). AI washing will influence market expectations, and Tobin's Q can directly capture the potential impact of this behavior. Tobin's Q is calculated as a firm's market value divided by the book value of its assets (Saxena et al., 2025; Guo et al., 2026).

3.2.2. Independent variable: AI washing (AIWashing)

We define AI washing by quantifying the gap between what firms say and what they do (Song et al., 2026; Sun et al., 2026). Following Xu et al. (2019) and Chen et al. (2025), we measure AI washing via the residual method:

$$AISymAct_{i,t} = \alpha_0 + \alpha_1 AISubAct_{i,t} + \sum \alpha_k Controls_{i,t} + \mu_i + \theta_t + \varepsilon_{i,t} \quad (1)$$

where, i and t denote the firm and year, respectively.$AISymAct_{i,t}$represents AI claims ("say"), measured by the logarithm of AI word frequency in annual reports.$AISubAct_{i,t}$ represents actual AI action ("do"). Consistent with Zhou et al. (2026), we measure actual AI action as the logarithm of total investment in AI hardware and software. The above data were all downloaded from the "Artificial Intelligence" section of the "Corporate Research Series" sub-database in the CSMAR database.

$Controls_{i,t}$ is a set of control variables (Details are provided below). Firm(μ_i) and year (θ_t) fixed effects are incorporated. The residual obtained from Model (1) is the degree of AI washing (*AIWashing*).

3.2.3. Moderators

Consistent with prior literature, firm size (*Size*) is measured as the natural logarithm of total assets. For state ownership (*SOE*), we employ a dummy variable, which equals 1 if the firm is ultimately controlled by government agencies, and 0 otherwise. High-tech enterprise (*TechInd*) is measured by a dummy variable indicating whether the firm operates in a high-tech industry.

3.2.4. Control variables

We control for the following variables: firm age (*FirmAge*), leverage (*Lev*), firm size (*Size*), institutional investor shareholding (*InsInvestor*), ownership concentration (*LarHolder*), management shareholding (*MShare*), state ownership (*SOE*), R&D subsidy (*RDsubsidy*), financing constraint (*SA*), R&D investment (*RDPay*), and high-tech industry (*TechInd*), consistent with the control variables in Model (1). In addition, firm and year fixed effects are adopted.

3.3. Model specification

To empirically explore the impact of AI washing on firm value, the following model is specified:

$$TQ_{i,t} = \beta_0 + \beta_1 AIWashing_{i,t} + \beta_2 AIWashing_{i,t}^2 + \sum \beta_k Controls_{i,t} + \mu_i + \theta_t + \varepsilon_{i,t} \quad (2)$$

Where, i and t denote the firm and year, respectively; $TQ_{i,t}$is firm value; $AIWashing_{i,t}$ is the independent variable; $AIWashing_{i,t}^2$ is the squared term of AI washing; $Controls_{i,t}$ is the set of control variables; $\mu_i$and$\theta_t$represent firm fixed effects and year fixed effects, respectively; $\varepsilon_{i,t}$ is the random error term. Table 1 presents the descriptive statistics.

Table 1
Descriptive statistics.

VarName	Obs	Mean	SD	Min	Median	Max
TQ	25,334	1.913	1.162	0.850	1.545	8.119
AIWashing	25,334	−0.000	0.468	−2.768	−0.004	3.094
AIWashing^2	25,334	0.219	0.428	0.000	0.064	9.575
FirmAge	25,334	20.930	6.075	3.000	21.000	36.000
Lev	25,334	0.402	0.202	0.056	0.391	0.975
Size	25,334	22.276	1.311	18.875	22.053	27.074
InsInvestor	25,334	40.823	25.312	0.000	40.730	92.317
LarHolder	25,334	32.651	14.571	8.410	30.220	75.000
MShare	25,334	0.097	0.152	0.000	0.009	0.601
SOE	25,334	0.338	0.473	0.000	0.000	1.000
RDsubsidy	25,334	9.633	7.062	0.000	13.592	17.743
SA	25,334	−3.920	0.264	−4.523	−3.918	−2.822
RDPay	25,334	18.067	1.474	13.756	18.045	21.946
TechInd	25,334	0.679	0.467	0.000	1.000	1.000

4. Empirical analysis

4.1. Basic regression analysis

As shown in Column (1) of Table 2, the coefficient of *AIWashing* is 0.026 ($p < 0.05$), while the coefficient of the squared term is −0.033 ($p < 0.1$). Further tests show that the slope at the lower bound of the data range is 0.210 ($p < 0.1$), whereas the slope at the upper bound is −0.180 ($p < 0.05$). The turning point of the curve is 0.392. The 95% confidence interval, calculated using the FIELLER method, is [0.113, 1.590], which falls well within the data range of *AIWashing* [−2.768, 3.094]. Collectively, these results provide robust evidence of a significant inverted U-shaped relationship between AI washing and firm value, supporting Hypothesis 1.

Column (2) shows that the coefficient of $AIWashing^2 \times Size$ is significantly negative at the 5% level, supporting Hypothesis 2. Column (3) shows that the coefficient of $AIWashing^2 \times SOE$ is significantly negative at the 10% level, supporting Hypothesis 3. Column (4) shows that the coefficient of $AIWashing^2 \times TechInd$ is significantly negative at the 10% level, supporting Hypothesis 4.

4.2. Endogeneity test

This study may be subject to two endogeneity concerns. First, reverse causality may exist: firm value could inversely affect its motivation and extent of AI washing. Second, omitted variable bias may exist.

We employ the regulatory distance from a firm's office to the provincial Securities Regulatory Bureau as an instrumental variable (*IVA*) (Zhong et al., 2025; Du et al., 2026). Regarding exogeneity, geographic distance is clearly exogenous and does not directly affect firm value. Regarding relevance, the greater the distance from the local Securities Regulatory Bureau, the weaker the regulatory oversight. This, in turn, increases the level of AI washing. Since this variable is time-invariant, we interact it with year dummies and use the interaction term as the instrumental variable (*IVA*). The first-stage results in Table 3(1) and (2) satisfy the relevance condition. The Cragg-Donald Wald F statistic (16.427) exceeds the critical value, meeting the weak IV test. The second-stage results of Column (3) shows that the coefficient of *AIWashing* is positive and significant at 10%, and $AIWashing^2$ is negative and significant at 10%. The *U* test confirms that the inverted U-shaped relationship remains significant, consistent with the baseline regression.

4.3. Robustness checks

We employ six alternative measures for AI washing. **First**, we replaced the actual AI action ($AISubAct_{i,t}$) with the ratio of total AI investment to total assets in Model (1), then re-estimated the regression to obtain the residual, which we labeled as *AIWashingA*. **Second**, we replaced the symbolic AI claims ($AISubAct_{i,t}$) with the logarithm of AI word frequency specifically within the MD&A section in Model (1), yielding *AIWashingB*. **Third**, we re-estimated Model (1) using the two alternative proxies adopted above, and

Table 2
Basic regression results.

Variables	(1)	(2)	(3)	(4)
AIWashing	**0.026****	0.401**	0.034**	0.029*
	(0.011)	(0.192)	(0.014)	(0.017)
$AIWashing^2$	**−0.033***	0.593*	−0.014	0.006
	(0.017)	(0.321)	(0.023)	(0.028)
AIWashing × Size		−0.017**		
		(0.008)		
$AIWashing^2 \times Size$		**−0.028****		
		(0.014)		
AIWashing × SOE			−0.022	
			(0.021)	
$AIWashing^2 \times SOE$			**−0.054***	
			(0.031)	
AIWashing × TechInd				−0.005
				(0.022)
$AIWashing^2 \times TechInd$				**−0.063***
				(0.034)
Size	−0.836***	−0.826***	−0.837***	−0.835***
	(0.035)	(0.035)	(0.035)	(0.035)
SOE	−0.170***	−0.170***	−0.156***	−0.170***
	(0.047)	(0.047)	(0.048)	(0.047)
TechInd	0.004	0.005	0.005	0.024
	(0.053)	(0.053)	(0.053)	(0.053)
Controls	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
Firm FE	Yes	Yes	Yes	Yes
Observations	25,334	25,334	25,334	25,334
R^2	0.726	0.726	0.726	0.726

Note: ***1%, **5%, *10%; the standard errors were clustered at the firm level.

Table 3
Regulatory distance as the instrumental variable.

Variables	First Stage		Second Stage
	AIWashing	$AIWashing^2$	TQ
IVA	0.029***		
	(0.002)		
IVA^2		0.027**	
		(0.012)	
AIWashing			0.070*
			(0.041)
$AIWashing^2$			−1.066*
			(0.564)
Controls	Yes	Yes	Yes
Year FE	Yes	Yes	Yes
Firm FE	Yes	Yes	Yes
Observations	25,334	25,334	25,334
Cragg-Donald Wald F statistic	16.427		

Note: ***1%, **5%, *10%; the standard errors were clustered at the firm level.

labeled the resulting residual *AIWashingC*. **Fourth**, we re-estimated the regression after removing the control variables from Model (1) and labeled the resulting residual *AIWashingD*. **Fifth**, we also used the log-transformed number of AI patents to measure firms' actual AI action, replaced the AI investment in Model (1) with it, and labeled the resulting residual *AIWashingE*.

Sixth, following Sun et al. (2026), we re-measure AI washing using Model (3).

$$AIwashingF_{i,t} = \frac{AISymAct_{i,t} - \overline{AISymAct_t}}{\sigma AISymAct} - \frac{AISubAct_{i,t} - \overline{AISubAct_t}}{\sigma AISubAct} \tag{3}$$

Where, $\overline{AISymAct_t}$ and $\overline{AISubAct_t}$ denote the cross-sectional averages of AI keyword frequency and AI investment for all firms in year *t*, respectively. $\sigma AISymAct$ and $\sigma AISubAct$ refer to the standard deviations across all firms in year *t*.

As shown in Table 4, regression results remain consistent with the baseline findings. The *U* test results indicate that the inverted U-shaped relationship is significant at the 5% level (Column 1), 1% level (Column 2), 5% level (Column 3), 5% level (Column 4), 5% level (Column 5) and 1% level (Column 6), respectively.

Table 4
Alternative measures for AI washing.

Variables	(1)	(2)	(3)	(4)	(5)	(6)
AIWashingA	0.025**					
	(0.011)					
$AIWashingA^2$	−0.032*					
	(0.017)					
AIWashingB		0.043***				
		(0.010)				
$AIWashingB^2$		−0.042***				
		(0.014)				
AIWashingC			0.042***			
			(0.010)			
$AIWashingC^2$			−0.041***			
			(0.014)			
AIWashingD				0.025**		
				(0.011)		
$AIWashingD^2$				−0.044***		
				(0.017)		
AIWashingE					0.026**	
					(0.011)	
$AIWashingE^2$					−0.040**	
					(0.017)	
AIWashingF						0.018*
						(0.011)
$AIWashingF^2$						−0.014***
						(0.005)
Controls	Yes	Yes	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes	Yes	Yes
Firm FE	Yes	Yes	Yes	Yes	Yes	Yes
Observations	25,334	25,334	25,334	25,334	25,334	25,334
R^2	0.726	0.726	0.726	0.726	0.726	0.726

Note: ***1%, **5%, *10%; the standard errors were clustered at the firm level.

5. Conclusion

This study finds a significant inverted U-shaped relationship between AI washing and firm value. Furthermore, this relationship is more pronounced in large, state-owned, and high-tech enterprises. The results have some practical implications.

First, our findings alert managers that although moderate AI washing may allow firms to gain temporary benefits from the market's enthusiasm for AI, this does not mean the market will passively and indefinitely accept unverified claims. In fact, excessive AI washing, while it captures market attention, also invites stricter scrutiny. Once a gap between a firm's claims and its actual capabilities is detected, trust erodes rapidly and results in substantial penalties on firm value.

Second, investors should pay particular attention to the AI claims of large, state-owned, and high-tech enterprises, as they may face more severe market penalties. A prudent approach would involve systematically verifying disclosed AI application scenarios, investment intensity, and revenue contributions, thereby distinguishing genuine capability signals from opportunistic hype.

Third, regulators should formulate specific AI information disclosure guidelines that encourage or require firms to disclose concrete AI application scenarios, investment amounts, output benefits, and risk assessments. Such norms would reduce information asymmetry, help stakeholders to distinguish credible capability signals from empty claims.

CRediT authorship contribution statement

Licheng Yang: Writing – review & editing, Writing – original draft, Methodology, Funding acquisition, Formal analysis, Data curation, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgements

This work was supported by the National Natural Science Foundation of China [grant no 72572081] and the Fundamental Research Funds for the Central Universities [grant no 60201525008]

Data availability

Data will be made available on request.

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